# TASK 2 – DATA ANALYSIS AND VISUALIZATION

## 1. Introduction

Data analysis and visualization help transform raw data into meaningful information. In this task, the USGS earthquake dataset was analysed using Microsoft Excel to identify patterns in earthquake frequency, magnitude, and depth.

## 2. Objective

The main objectives of this task are:

- To analyse earthquake events based on their occurrence month.
- To classify earthquakes according to magnitude.
- To study earthquake depth categories.
- To identify patterns and trends in the dataset.
- To present the findings through charts and an interactive dashboard.

## 3. Tools Used

- Microsoft Excel
- PivotTables
- PivotCharts
- Excel formulas
- Slicers

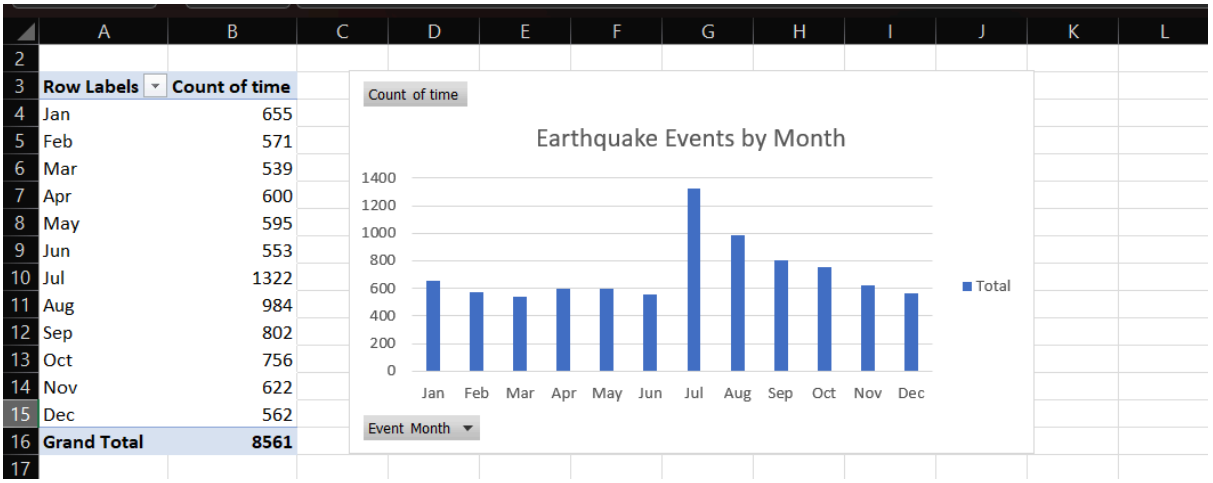
## 4. Data Analysis

The cleaned dataset containing 8,561 earthquake records was used for analysis.

### 4.1 Monthly Earthquake Analysis

A PivotTable was created to calculate the number of earthquake events recorded in each month.

The analysis showed that July had the highest number of earthquake events, with 1,322 records.

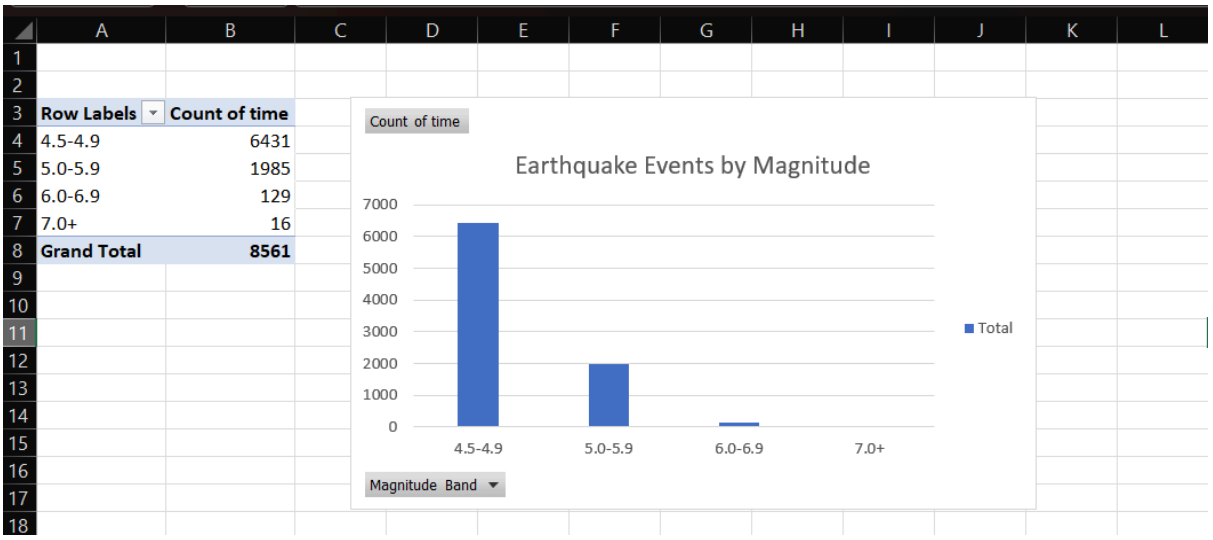



## 4.2 Magnitude Analysis

Earthquake events were classified into four magnitude groups:

| Magnitude Band | Number of Events |
|---|---|
| 4.5–4.9 | 6,431 |
| 5.0–5.9 | 1,985 |
| 6.0–6.9 | 129 |
| 7.0+ | 16 |
| **Total** | **8,561** |

The results show that most recorded events belonged to the 4.5–4.9 magnitude group.



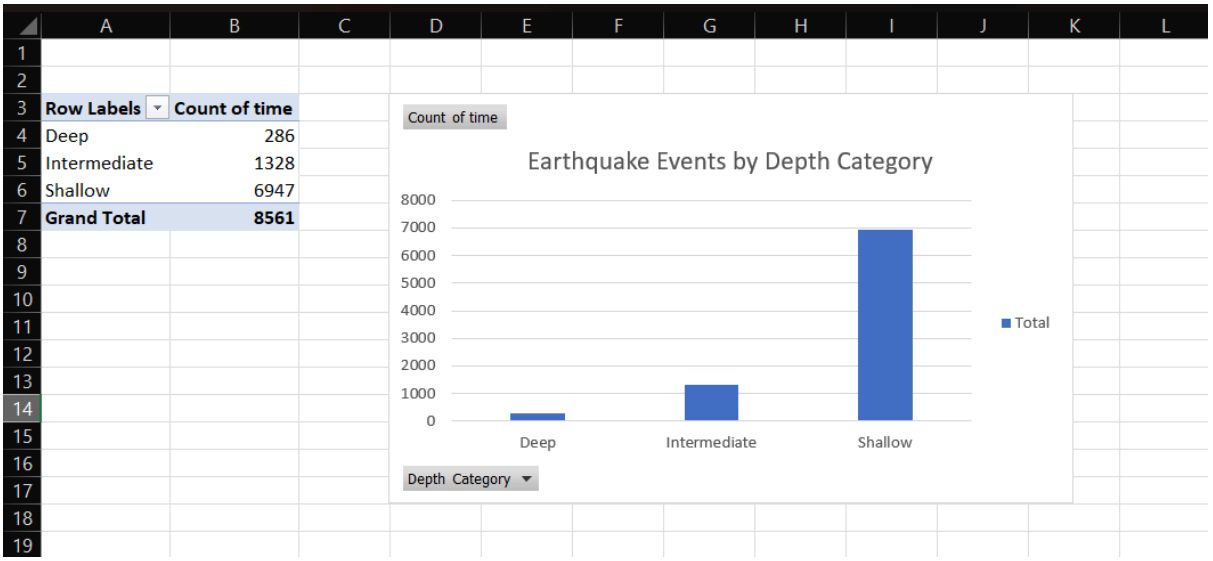

## 4.3 Depth Analysis

Earthquakes were classified into three categories based on depth:

- Shallow: Below 70 km

- Intermediate: 70 km to below 300 km
- Deep: 300 km and above

A PivotTable and PivotChart were used to compare the number of events in each category.

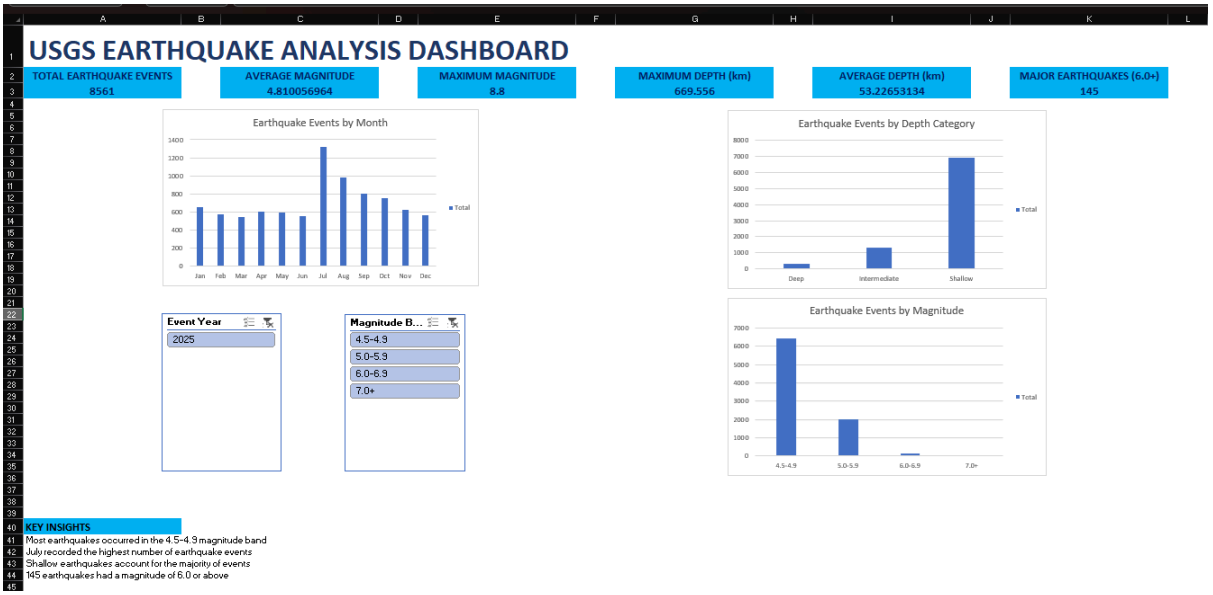



# 5. Dashboard Development

An interactive Excel dashboard was created to present the analysis in a simple and understandable format.

The dashboard includes:

- Total earthquake events
- Average magnitude
- Maximum magnitude
- Maximum depth
- Average depth
- Number of major earthquakes (magnitude 6.0 and above)
- Monthly earthquake chart
- Magnitude distribution chart
- Depth category chart
- Magnitude Band and Event Year slicers




# 6. Key Findings

1. A total of 8,561 earthquake records were analysed.
2. July recorded the highest number of events, with 1,322.
3. The 4.5–4.9 magnitude category contained the largest number of events.
4. A total of 145 earthquakes had a magnitude of 6.0 or above.
5. Interactive slicers allow users to filter and explore the dashboard.

# 7. Conclusion

This task helped analyse earthquake data and identify useful patterns related to monthly occurrence, magnitude, and depth. Microsoft Excel PivotTables, PivotCharts, formulas, and slicers were used to create an interactive dashboard. The final dashboard presents the findings in a clear and organised manner and makes the dataset easier to understand.